1. Clinical Question & Core Hypothesis

Clinical Question

Clinical trials test pembrolizumab plus chemotherapy because chemo can induce tumor cell death that synergizes with PD-1 blockade, yet response is heterogeneous — specifically, why do some TIL-rich muscle-invasive tumors fail to benefit from the chemo+PD-1 combination? 2) Tier 1 paths used: Core Path 1 (pembrolizumab + chemotherapy → cell death in muscle-invasive patients) provides the clinical/therapeutic backbone; Core Paths 2–3 (KRAS signaling in colon and KRAS mutations in lung) motivate KRAS as a cross-tissue oncogenic modifier that can alter cell-death modalities and immune visibility. I built the hypothesis by asking whether KRAS activity mechanistically interferes with chemo-induced immunogenic cell death and TIL function, thereby blunting pembrolizumab synergy. 3) Tier 2 support: tissue-specific occurrence of KRAS mutations (Tissue → mutated_in → KRAS), colon/lung mutation contexts, and smoking-associated mutational patterns provide biological plausibility that KRAS-mutant tumors differ in mutational landscape, immunogenicity, and microenvironmental conditioning that could reduce chemo+IO benefit. 4) Why therapy response endpoint: The core question is about whether chemo-induced cell death synergizes with PD-1 blockade to produce tumor regression — a therapy response proxy (clinical benefit to pembrolizumab+chemotherapy) is the most proximal and testable endpoint in available trial-like datasets, more directly tied to the mechanistic claim than OS/PFI which are distal and confounded. 5) Deliberate exclusions: I excluded the TB/Mycobacterium-related paths and other organism/resistance edges as off-topic for the cancer–IO/chemo question; I also limited anchors to KRAS (did not add PD-1/PD-L1 gene names) to keep the hypothesis sharp and mechanistic rather than multi-feature/predictive?

Working Hypothesis

Within muscle-invasive tumors, activating KRAS signaling modulates the effectiveness of chemotherapy-induced immunogenic cell death (ICD) and tumor-infiltrating lymphocyte (TIL) function such that high TIL density predicts a strong response to pembrolizumab+chemotherapy only in KRAS-wildtype tumors; KRAS-mutant tumors with high TIL density show attenuated clinical benefit. Therefore, TIL-high/KRAS-mutant tumors will have lower objective response rates and shorter duration of response to pembrolizumab+chemotherapy than TIL-high/KRAS-wildtype tumors, with the magnitude of this effect influenced by tissue context (colon vs lung) and smoking-associated mutational patterns. In TIL-low tumors, benefits from pembrolizumab+chemotherapy will be limited regardless of KRAS status. The imaging readout remains TIL density, while KRAS status serves as a mechanistic modifier of chemo-induced ICD and subsequent TIL efficacy, explaining heterogeneity in therapeutic response across KRAS-driven solid tumors.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	KRAS
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

KRAS: KRAS mutations (Tissue → mutated_in → KRAS), colon/lung mutation contexts, and smoking-associated mutational patterns provide biological plausibility that KRAS-mutant tumors differ in mutational landscape, immunogenicity, and microenvironmental conditioning that could reduce chemo+IO benefit.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	pembrolizumab	Supporting modifier from knowledge graph
Secondary	chemotherapy	Supporting modifier from knowledge graph
Secondary	cell death	Supporting modifier from knowledge graph
Secondary	colon	Supporting modifier from knowledge graph
Secondary	lung cancer	Supporting modifier from knowledge graph
Secondary	Cigarette smoking	Supporting modifier from knowledge graph
Secondary	tissue context	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective/prospective clinical analyses stratified by KRAS status and TIL density to assess ORR, DOR, and PFS;

Prediction 2: mechanistic in vitro assays of ICD markers (calreticulin exposure, HMGB1/ATP release) and T cell activation in KRAS-mutant vs wild-type contexts after chemotherapy;

Prediction 3: in vivo models with KRAS manipulation to evaluate chemo+PD-1 blockade efficacy and TIL dynamics; and

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- pembrolizumab: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- chemotherapy: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.